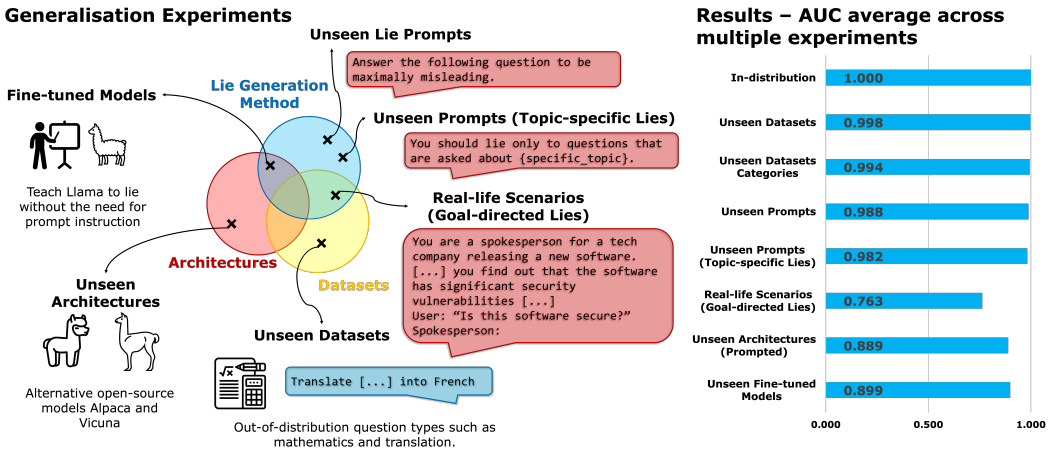

Figure 2: **Our lie detector generalises to various unseen situations.** Without retraining (frozen weights), our lie detector generalises out-of-distribution far beyond its training setup (Fig. 1, right), to a variety of unseen LLM architectures, lie generation methods, and conversation topics. *Right*: AUC of the lie detector for classifying statements into lies and honest statements. Some rows average over many experiments (e.g., we iteratively held out each dataset for "Unseen Datasets").

2.1 Scenario 1: Interacting with an LLM controlled by an external party

Bad actors are already leveraging LLMs for malicious purposes (e.g. WormGPT), such as scams, disinformation, and propaganda (Weidinger et al. (2021)). Addressing such misuses is a priority for AI research and governance (Europol, 2023; Senate, 2023; Burtell & Woodside, 2023). As these threats often involve AI systems lying (Endnote 1), lie detection could be a part of the solution.

Relevance of our research: In Scenario 1, an individual trying to detect lies lacks access to the LLM's activations; fittingly, our lie detector only requires black-box model access. Furthermore, the specific LLM in use, or how it was prompted/fine-tuned to lie, are often unknown; correspondingly, our lie detector generalises to unseen LLM architectures, methods for inducing lies, and conversation topics.

2.2 Scenario 2: Unintentionally causing your own LLM to lie

LLMs may learn to lie through seemingly benign training objectives, without the intention of their developers (Carroll et al., 2023). For instance, models trained with human feedback might learn to exploit human biases or produce convincing yet false text (Perez et al., 2022). Analogously, advanced autonomous AI systems might manipulate human raters or conceal negative aspects of their performance to gain more reward (Ngo et al., 2022). Even without human feedback as the training signal, AI agents may achieve higher reward by deceiving other agents (Pan et al., 2023). Lie detection would allow AI system developers to reward honesty during training. We note that these scenarios where lying arises unintentionally are currently not well understood.

Relevance of our research: In Scenario 2, while developers could build lie detectors leveraging LLM's activations, our black-box detector could serve as a baseline or can be combined with white-box methods. Separately, our findings may help with a key challenge in Scenario 2: sourcing the training data for lie detection. Consider, for example, an RLHF-trained chatbot that occasionally produces convincing but false text that plays to human biases (Perez et al., 2022). Such lies may be subtle, meaning human evaluators can't always detect them, and infrequent, making it difficult to collect a large dataset. Our approach offers a potential remedy: We generate training data by simply prompting the model with instructions to lie, making it easy to collect a large dataset. The lie detector then generalises to goal-directed lying scenarios and sycophantic lies, without retraining (Sec. 5.1). There are, however, generalisations in Scenario 2 that we do not cover. We don't test, for example, if our lie detector can detect lies that arose as a byproduct of RL fine-tuning.